Fiscal Year:	FY 2008	**Task Last Updated:**	FY 01/18/2009
PI Name:	Gould, Harvey Allen Ph.D.		
Project Title:	Feasibility Study for an Electron Electric Dipole Moment		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	gould@lbl.gov	**Fax:**	FY 510-486-7981
PI Organization Type:	GOVERNMENT	**Phone:**	510-486-7777
Organization Name:	Lawrence Berkeley National Laboratory		
PI Address 1:	MS 71-259		
PI Address 2:	LBNL		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	09/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Maleki, Lute	**Contact Phone:**	818-354-3688
Contact Email:	lute.maleki@jpl.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/08, per PI and GRC (10/07) NOTE: Received NCE to 9/30/07, per PI and GRC (7/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	C73534 A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	To determine if a slow, laser-cooled atom microgravity experiment can improve the sensitivity to an electron electric dipole moment (e-EDM) by orders of magnitude. Develop and test experimental features such as electric field quantization, electrostatic beam focusing, use of field-free regions, and pulsed oscillatory fields. Determine sensitivity to systematic effects, sensitivity to electric and magnetic noise. Measure the Cs static dipole polarizability. Perform other tests and measurements incidental to these. Will determine the most useful features to incorporate, identify problem areas, and help assess the feasibility of proceeding to flight definition studies for a microgravity electron electric dipole moment experiment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Spinoff can improve accuracy of cesium atomic clocks in space and on Earth.
Task Progress:	A proof-of-principle electron electric dipole moment (e-EDM) experiment using slow cesium atoms, nulled magnetic fields, and electric field quantization has been completed, the data analyzed, and a manuscript published in Physical Review A78 063416 (2007). With the ambient magnetic fields seen by the atoms reduced to less than 200 pT, an electric field of 6 MV/m lifts the degeneracy between states of unequal mF and, along with the low (3 m/s) velocity, suppresses the systematic effect from the motional magnetic field. Preparation and planning for further improvements to electron EDM experiments has been carried out. This includes a comprehensive analysis of systematic effects, an analysis of magnetic shielding requirements and magnetic shield design, and work towards developing glass electric field plates that will operate at room temperature. Glass electric field plates sustain higher electric fields than metal plates (due to the absence of conducting electrons) and do not have the problem of magnetic Johnson/Nyquist noise from electron motion.
Bibliography Type:	Description: (Last Updated: 02/25/2014)